

DISTRIBUTED SYSTEMS AND CLOUD/EDGE COMPUTING
A.Y. 2024-2025, Prof. Claudio Savaglio

Smart Healthcare Monitoring System

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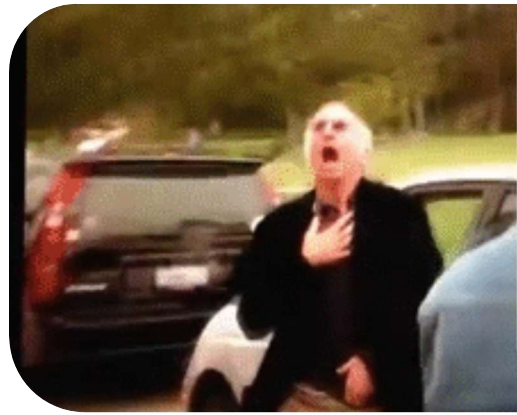
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DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

GOAL

HEART DISEASE
PREDICTION



PNEUMONIA DETECTION
FROM X-RAY



PERFORM HEALTH
MONITORING OPERATIONS



REQUIREMENT

FUNCTIONAL

- The system shall have a login and sign-up feature
- Model that can predict heart disease
- Model that can perform Pneumonia Detection on an X-ray image

REQUIREMENT

FUNCTIONAL

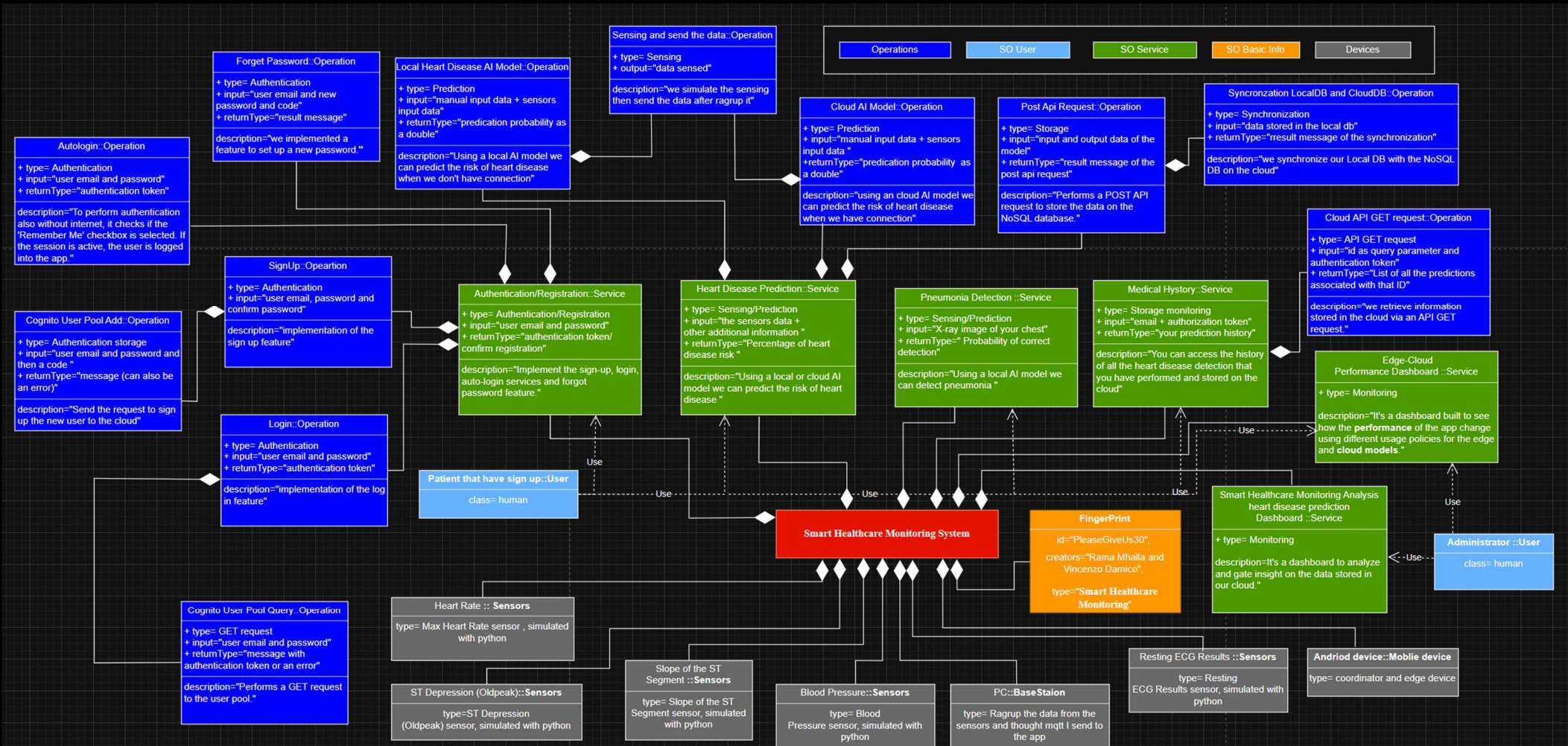
- Store the medical history related to the heart disease predictions
- Work without internet connection
- Analyze and retrieve insight about our system data and performance

REQUIREMENT

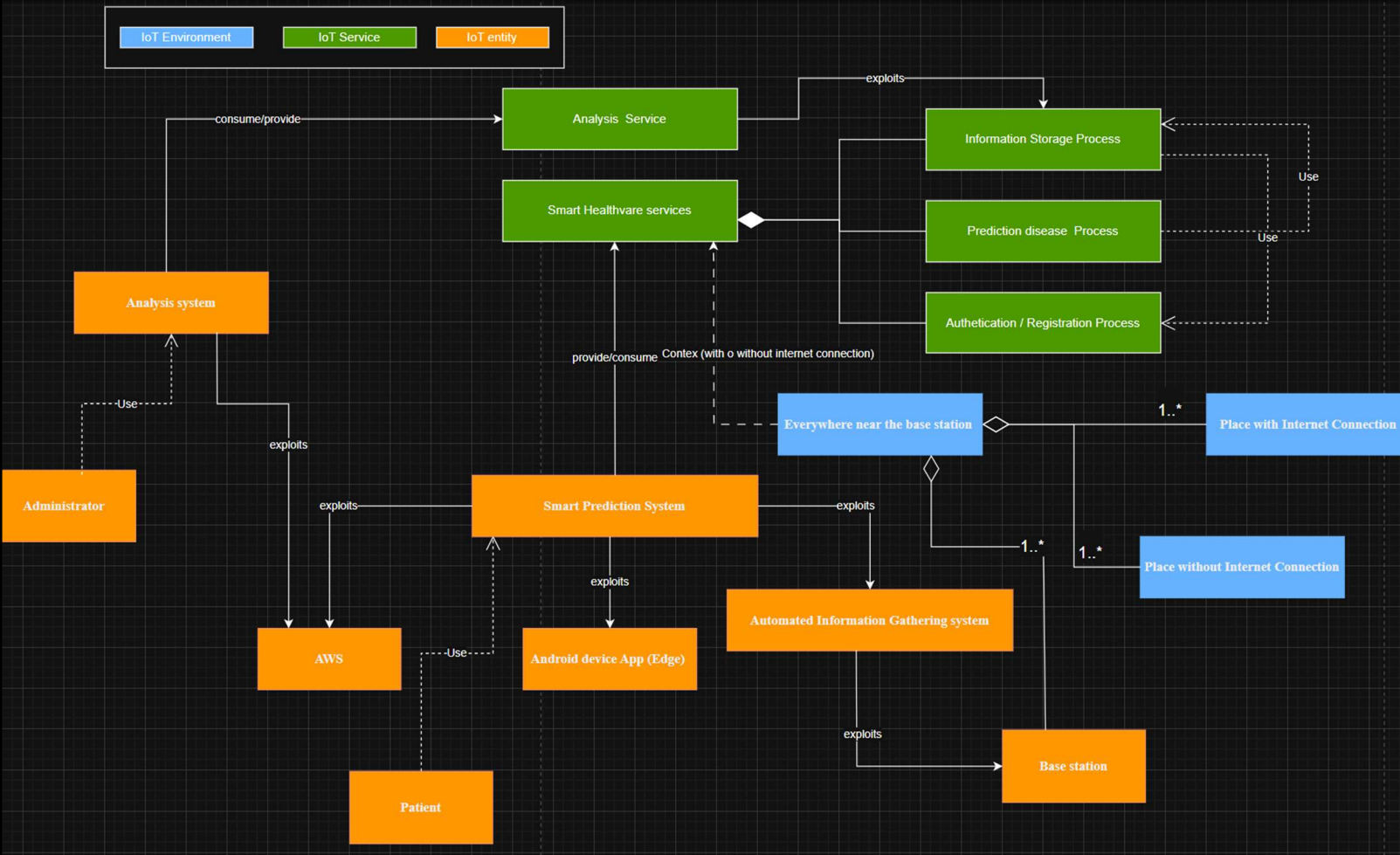
NON-FUNCTIONAL

- Manual and automatic mechanism to insert data
- Auto-login shall be performed even without an internet connection
- Heart disease model shall run on cloud and locally
- The app has to be fast and reliable

ACOSO-METH ANALYSIS MODEL



ACOSO-METH DOMAIN MODEL



Analysis Service

Services output	Data Insight Dashboard Performance Insight DashBoard
Services Input	Prediction records (Input + Output) / Simulations result

Smart Healthcare Services

Services output	Heart disease or X-ray prediction store and associate them with a user
Services Input	Sensors data or X-ray chest image + user credentials

Information Storage Process

Process Description	Store the heart disease detection that the user have performed and stored
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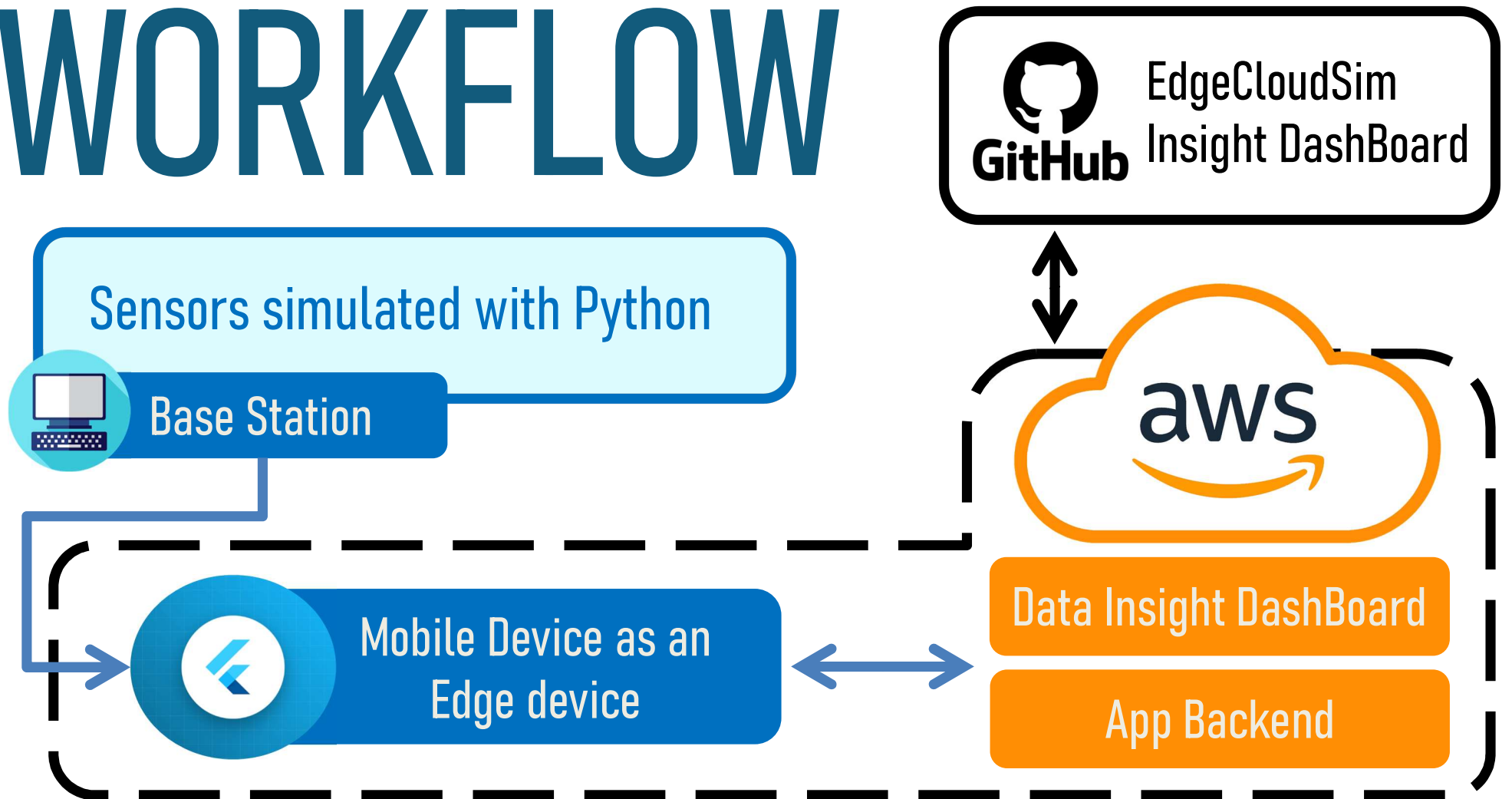
Authetication Registration Process

Process Description	Implement the sign-up, login, auto-login services and forgot password feature
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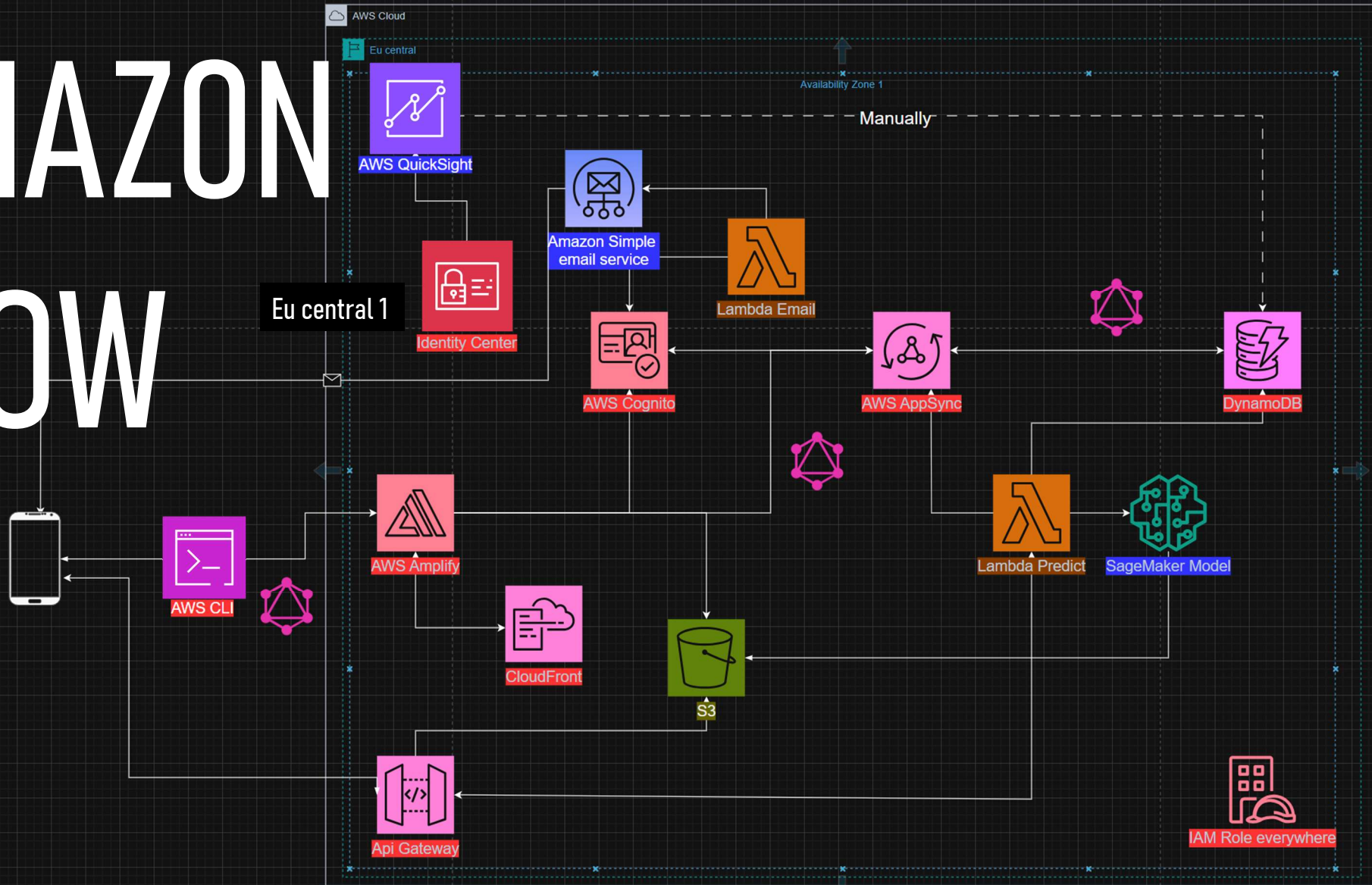
Prediction Disease Process

Process Description	The process can perform heart disease prediction and X-ray prediction
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WORKFLOW

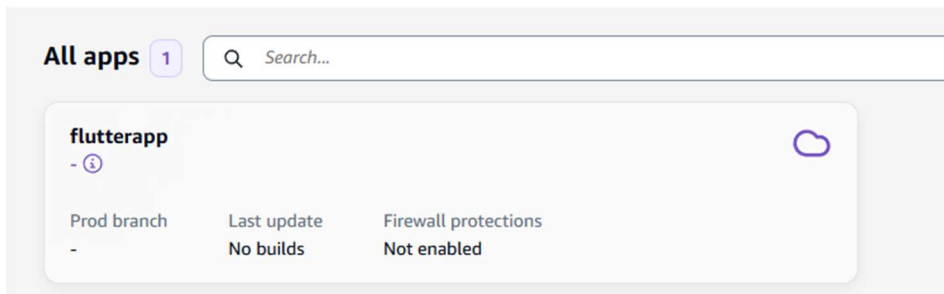


AMAZON FLOW

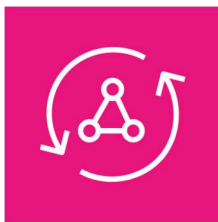




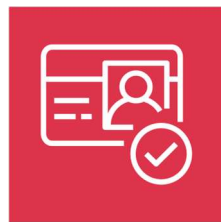
AWS Amplify



Dynamo



AppSync



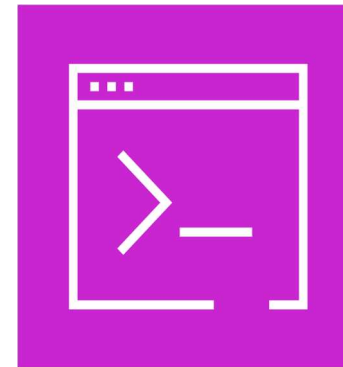
Cognito



S3

Configured

AWS CLI



IAM User





Notebook



Model



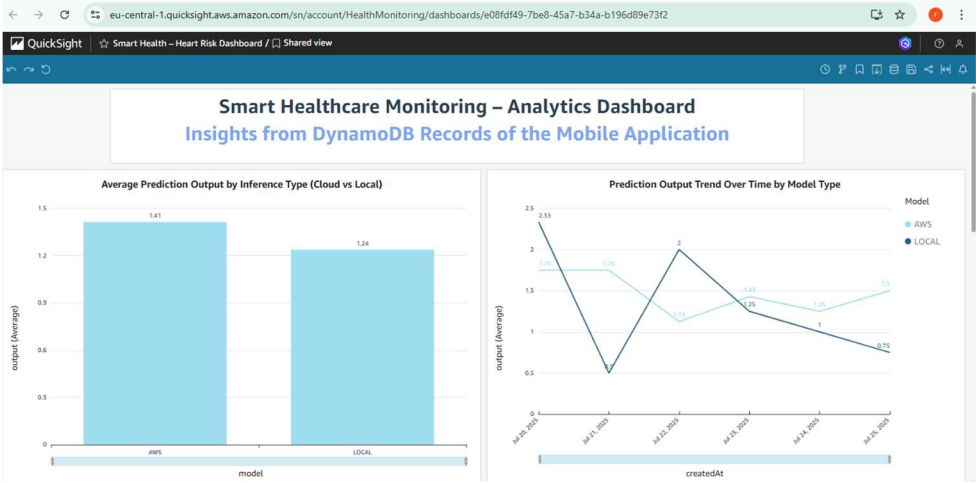
Train



Accuracy = 93.21%

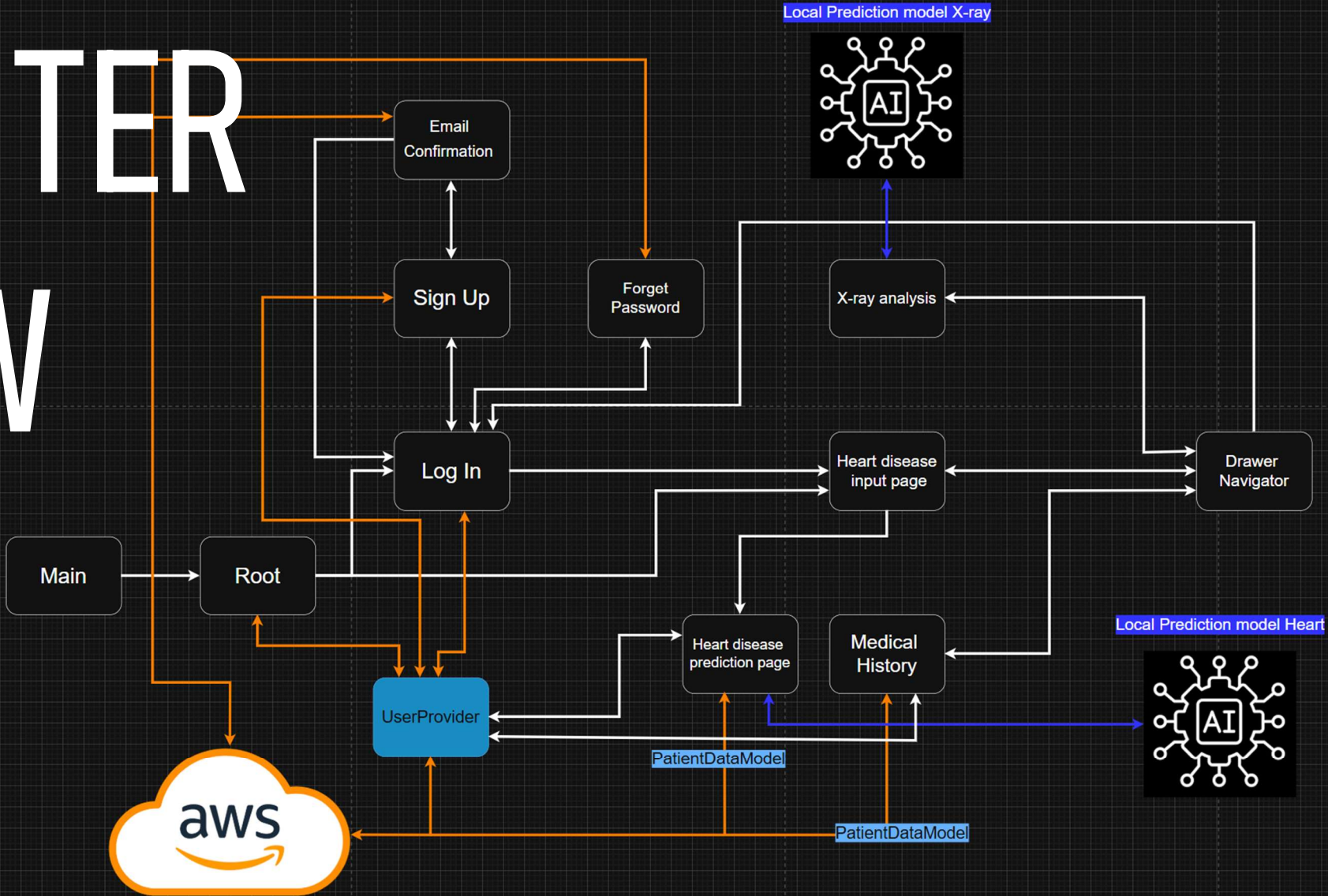
EdgeCloudSim Dashboard

	AI_BASED	RANDOM	PREDICTIVE	GAME_THEORY
Processing Time	⚡ Better (slightly lower)	⚡ Slightly better	⚡ Slightly better	⚡ Slightly better
Network Delay	⚡ Better (always lower)	⚡ A bit better	⚡ Slight edge	⚡ Slightly better
Failed Tasks (%)	⚡ More stable	📱 Better at some loads	⚡ Clearly better	📱 Slightly better at times
Service Time	⚡ Consistently lower	⚡ Slightly better	⚡ Slightly better	⚡ Consistently better



QuickSight Dashboard

FLUTTER FLOW



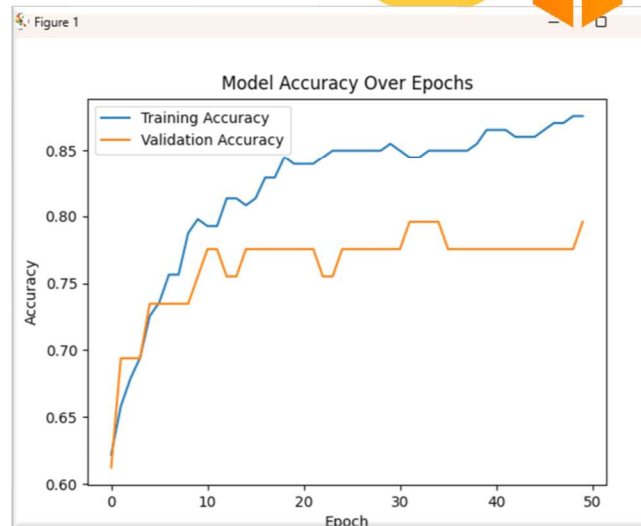
kaggle



LOCAL AI MODELS

Avg. latency = 19 ms

Accuracy = 85.25%



Sensor Data

Connected

Blood Pressure
187 mmHg

Resting ECG
0

Max Heart Rate
176 BPM

ST Depression
2.40

Slope
2

MQTT

Sensor Data

Disconnected

Enter Sensor Values Manually

Blood Pressure
120

Resting ECG
0-2

Max Heart Ra...
150

ST Depression
1.5

Slope
0-2

THANK YOU FOR YOUR ATTENTION!

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